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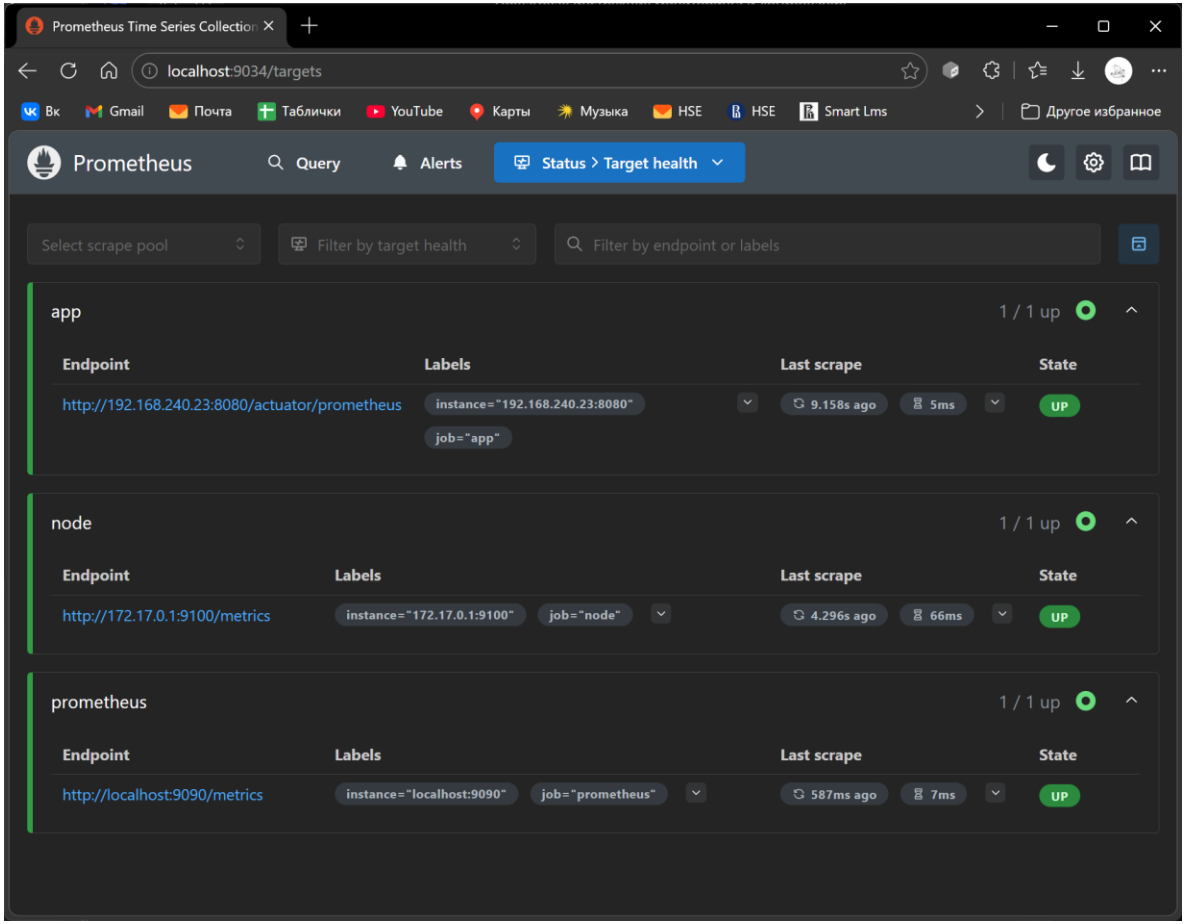
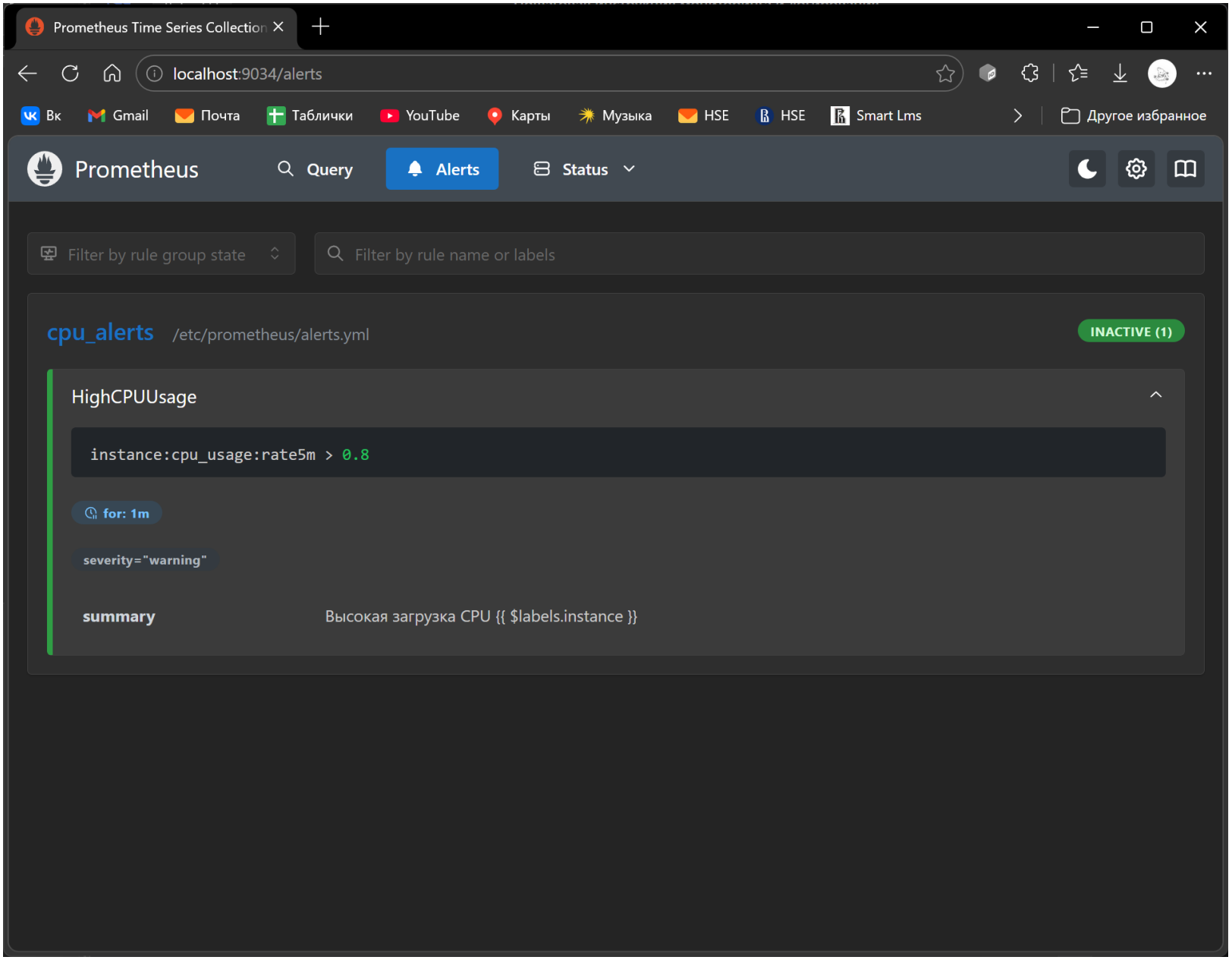
Практика 1: сбор метрик

```
red34@compute-vm-8-16-30-ssd-1772094666092:~$ docker exec -it prometheus_red34 sh -c "promtool query instant http://localhost:9090 'rate(node_disk_written_bytes_total[5m]) / 1024'"
{device="vda", instance="172.17.0.1:9100", job="node"} => 227.92391856974714 @[1774616400.524]
red34@compute-vm-8-16-30-ssd-1772094666092:~$ docker exec -it prometheus_red34 sh -c "promtool query instant http://localhost:9090 '(1 - (node_memory_MemAvailable_bytes / node_memo
ry_MemTotal_bytes)) * 100'"
{instance="172.17.0.1:9100", job="node"} => 20.428559633345756 @[1774616411.402]
red34@compute-vm-8-16-30-ssd-1772094666092:~$ docker exec -it prometheus_red34 sh -c "promtool query instant http://localhost:9090 '100 - (avg by(instance) (irate(node_cpu_seconds_
total{mode=\"idle\"}[5m])) * 100)'"
{instance="172.17.0.1:9100"} => 3.5416666666666714 @[1774616429.033]
red34@compute-vm-8-16-30-ssd-1772094666092:~$
```

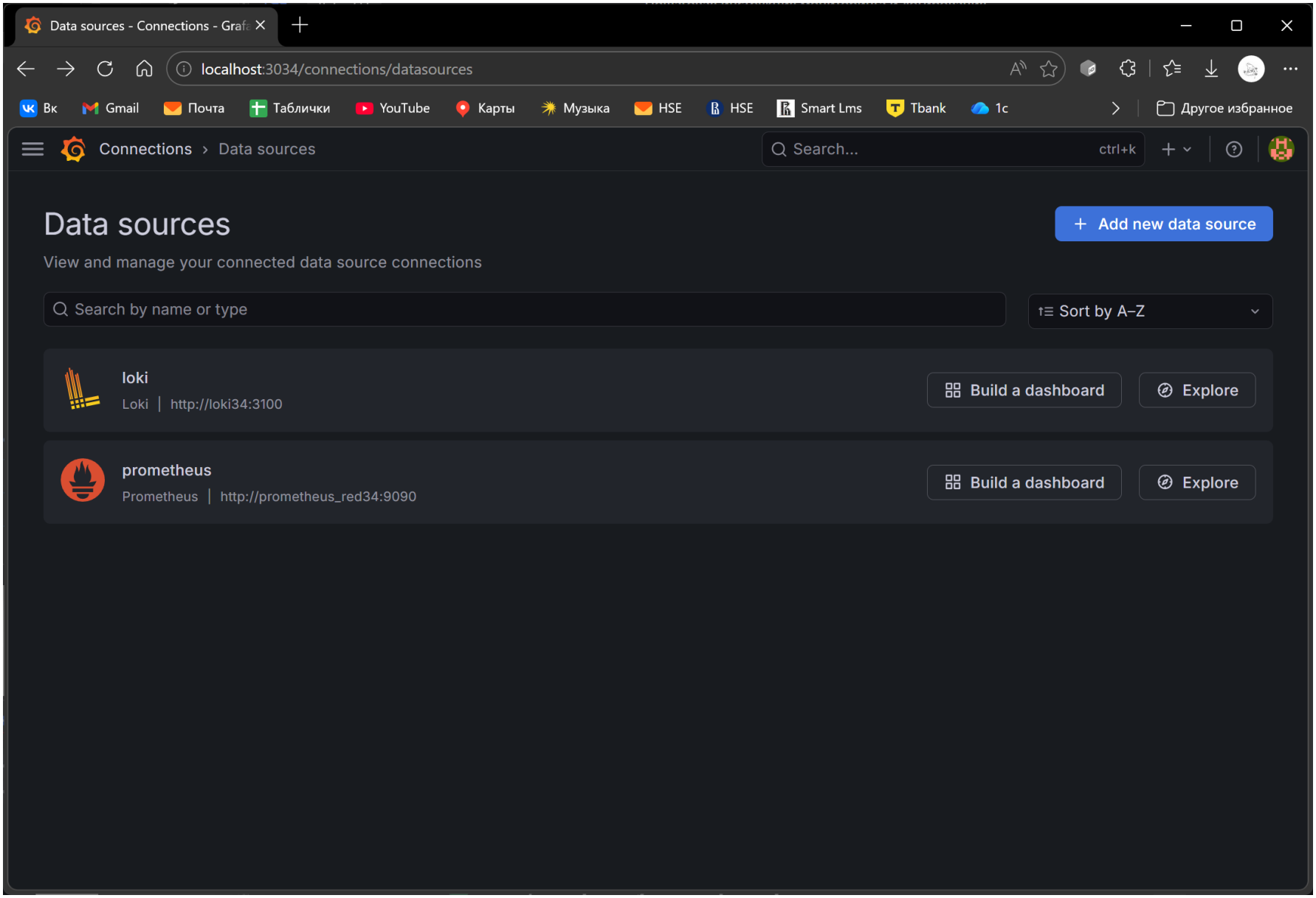
Практика 2: app

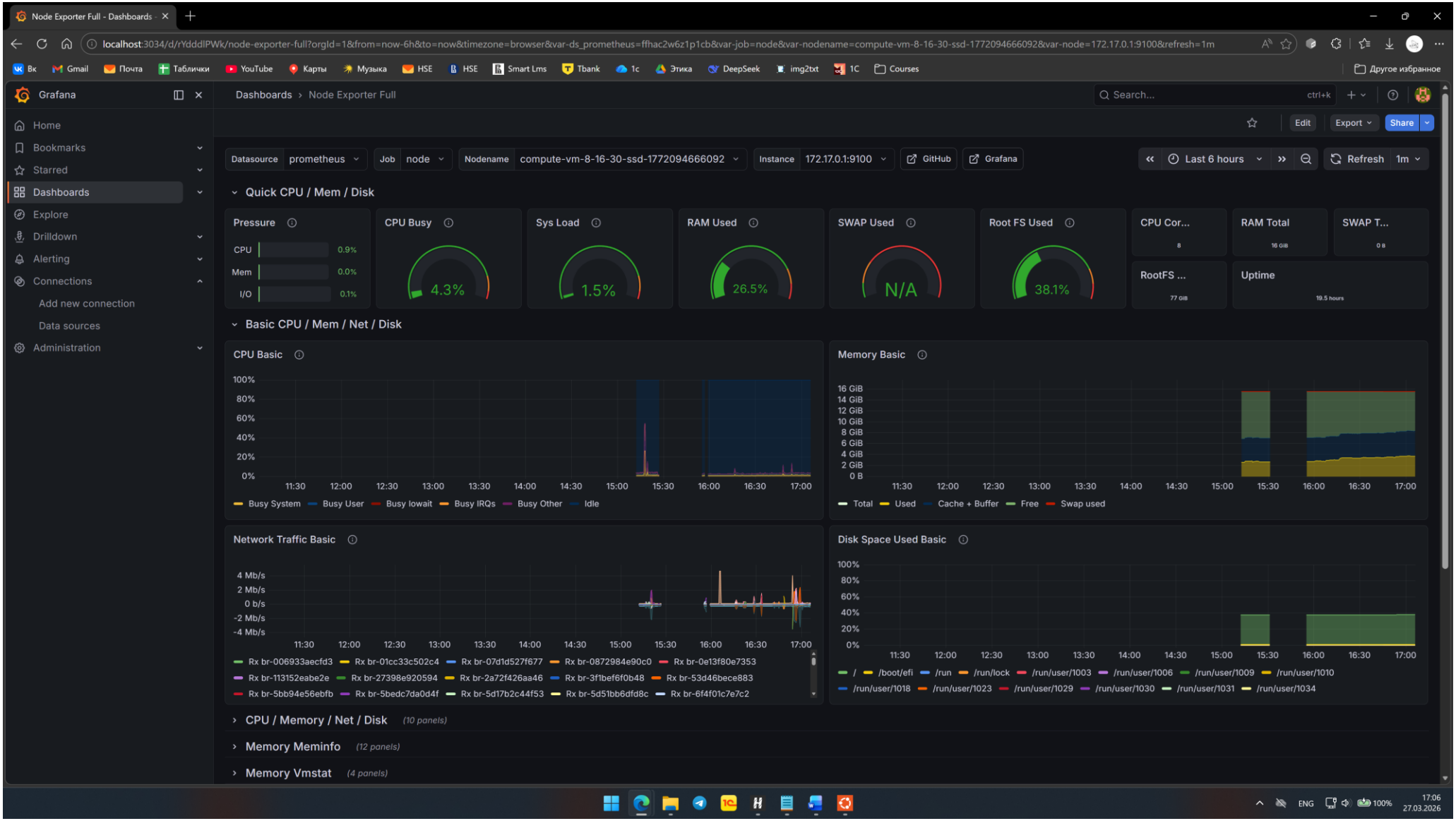


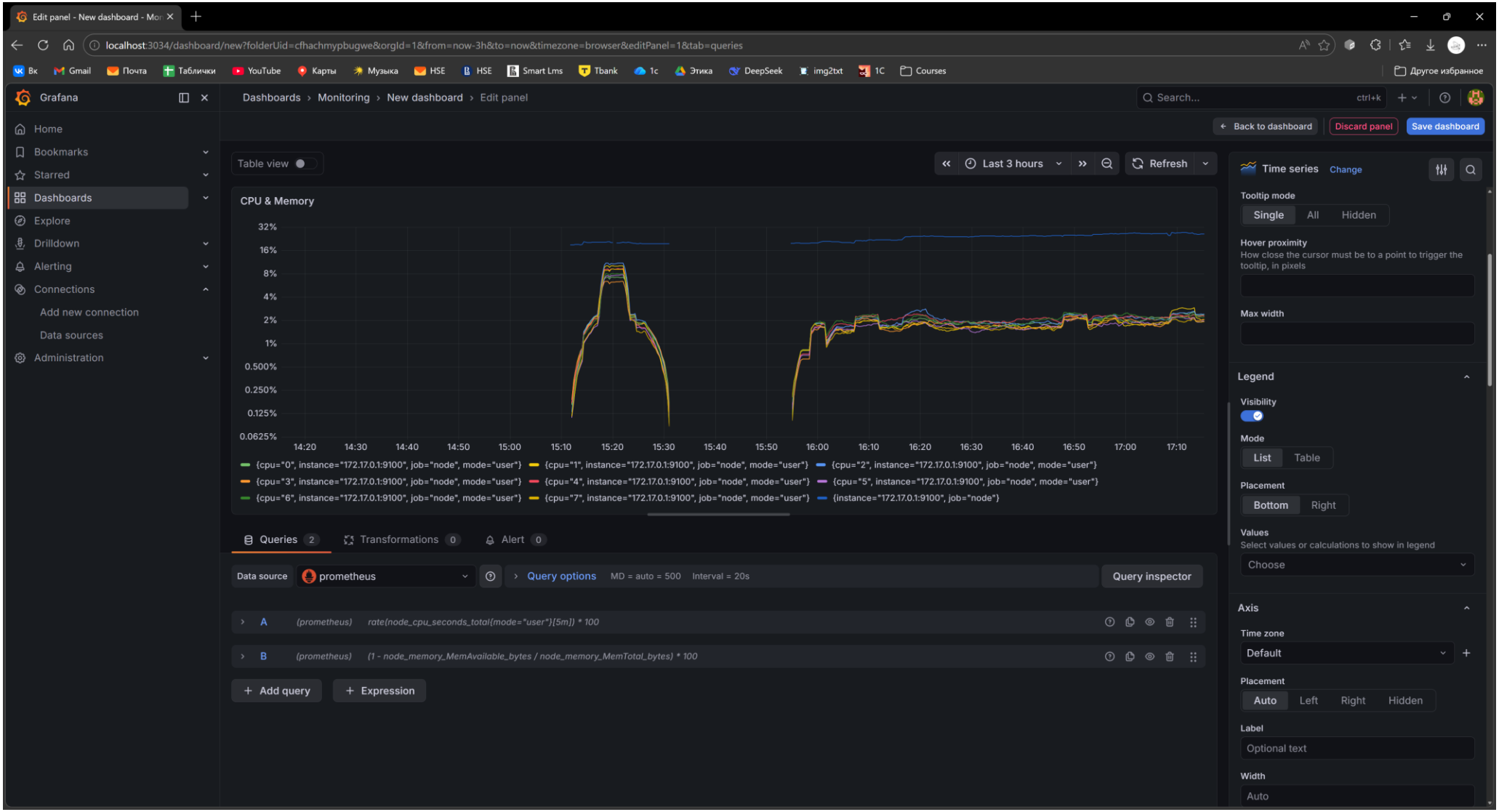
Практика 3: алерты



Практика 4:


Data sources - Connections - Grafa
localhost:3034/connections/datasources
Вк
Gmail
Почта
Таблички
YouTube
Карты
Музыка
HSE
HSE
Smart Lms
Tbank
1c
Другое избранное
Connections › Data sources
Search...
ctrl+k
Data sources
Add new data source
View and manage your connected data source connections
Search by name or type
Sort by A–Z
loki
Loki | http://loki34:3100
Build a dashboard
Explore
prometheus
Prometheus | http://prometheus_red34:9090
Build a dashboard
Explore


Node Exporter Full - Dashboards
localhost:3034/d/rYdddlPWk/node-exporter-full?orgId=1&from=now-6h&to=now&timezone=browser&var-ds_prometheus=ffhac2w6z1p1cb&var-job=node&var-nodename=compute-vm-8-16-30-ssd-1772094666092&var-node=172.17.0.1:9100&refresh=1m
Grafana
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Dashboards › Node Exporter Full
Search...
Edit
Export
Share
Datasource prometheus
Job node
Nodename compute-vm-8-16-30-ssd-1772094666092
Instance 172.17.0.1:9100
GitHub
Grafana
Last 6 hours
Refresh 1m
Quick CPU / Mem / Disk
Pressure
CPU 0.9%
Mem 0.0%
I/O 0.1%
CPU Busy
4.3%
Sys Load
1.5%
RAM Used
26.5%
SWAP Used
N/A
Root FS Used
38.1%
CPU Cor...
8
RAM Total
16 GiB
SWAP T...
0 B
RootFS ...
77 GiB
Uptime
19.5 hours
Basic CPU / Mem / Net / Disk
CPU Basic
Busy System
Busy User
Busy Iowait
Busy IRQs
Busy Other
Idle
Memory Basic
Total
Used
Cache + Buffer
Free
Swap used
Network Traffic Basic
Disk Space Used Basic
CPU / Memory / Net / Disk (10 panels)
Memory Meminfo (12 panels)
Memory Vmstat (4 panels)


Edit panel - New dashboard - Mon
localhost:3034/dashboard/new?folderUid=cfhachmypbugwe&orgId=1&from=now-3h&to=now&timezone=browser&editPanel=1&tab=queries
Grafana
Dashboards › Monitoring › New dashboard › Edit panel
Search...
Back to dashboard
Discard panel
Save dashboard
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Table view
Last 3 hours
Refresh
CPU & Memory
32%
16%
8%
4%
2%
1%
0.500%
0.250%
0.125%
0.0625%
14:20
14:30
14:40
14:50
15:00
15:10
15:20
15:30
15:40
15:50
16:00
16:10
16:20
16:30
16:40
16:50
17:00
17:10
{cpu="0", instance="172.17.0.1:9100", job="node", mode="user"}
{cpu="1", instance="172.17.0.1:9100", job="node", mode="user"}
{cpu="2", instance="172.17.0.1:9100", job="node", mode="user"}
{cpu="3", instance="172.17.0.1:9100", job="node", mode="user"}
{cpu="4", instance="172.17.0.1:9100", job="node", mode="user"}
{cpu="5", instance="172.17.0.1:9100", job="node", mode="user"}
{cpu="6", instance="172.17.0.1:9100", job="node", mode="user"}
{cpu="7", instance="172.17.0.1:9100", job="node", mode="user"}
{instance="172.17.0.1:9100", job="node"}
Queries 2
Transformations 0
Alert 0
Data source
prometheus
Query options
MD = auto = 500
Interval = 20s
Query inspector
A (prometheus) rate(node_cpu_seconds_total{mode="user"}[5m]) * 100
B (prometheus) (1 - node_memory_MemAvailable_bytes / node_memory_MemTotal_bytes) * 100
Add query
Expression
Time series
Change
Tooltip mode
Single
All
Hidden
Hover proximity
How close the cursor must be to a point to trigger the tooltip, in pixels
Max width
Legend
Visibility
Mode
List
Table
Placement
Bottom
Right
Values
Select values or calculations to show in legend
Choose
Axis
Time zone
Default
Placement
Auto
Left
Right
Hidden
Label
Optional text
Width
Auto

Edit panel - New dashboard - Dasl

localhost:3034/dashboard/new?orgId=1&from=now-6h&to=now&timezone=browser&editPanel=1

Grafana

Dashboards › New dashboard › Edit panel

Search... ctrl+k

Back to dashboard | Discard panel | Save dashboard

Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration

Table view

Last 6 hours | Refresh

RPC

0 req/s

Queries 1 | Transformations 0 | Alert 0

Data source: prometheus | Query options MD = auto = 500 Interval = 30s | Query inspector

A (prometheus)

Explain | Run queries | Builder | Code

Metrics browser ›

```
rate(http_server_requests_seconds_count{uri="/actuator/prometheus"}[5m])
```

1 `rate(http_server_requests_seconds_count{uri="/actuator/prometheus"}[5m])`
Fetch all series matching metric name and label filters.

2 `rate(<expr>[5m])`
Calculates the per-second average rate of increase of the time series in the range vector. Breaks in monotonicity (such as counter resets due to target restarts) are automatically adjusted for. Also, the calculation extrapolates to the ends of the time range, allowing for missed scrapes or imperfect alignment of scrape cycles with the range's time period.

Options Legend: Auto Format: Time series Step: auto Type: Range Exemplars: false

12.4 Stat Change

Calculate a single value per column or series or show each row

Calculate | All values

Calculation
Choose a reducer function / calculation
Last *

Fields
Select the fields that should be included in the panel
Numeric Fields

Stat styles

Orientation
Layout orientation
Auto | Horizontal | Vertical

Text mode
Control if name and value is displayed or just name
Auto

Color mode
Value

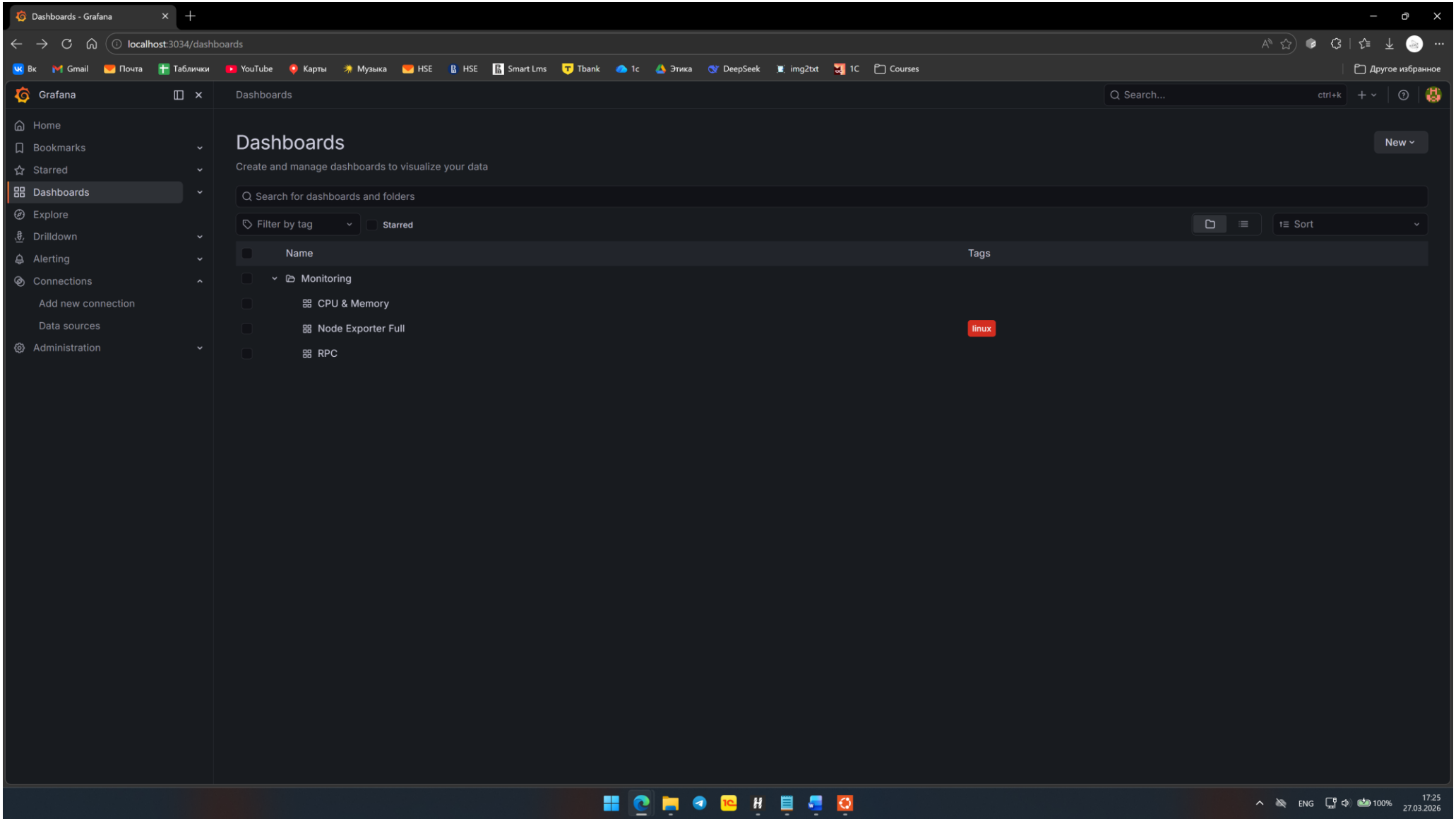
Graph mode
Stat panel graph / sparkline mode
None | Area

Text alignment
Auto | Center

Show percent change

Text size

Title
Auto


Dashboards - Grafana
localhost:3034/dashboards
Grafana
Dashboards
Search...
ctrl+k
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Dashboards
Create and manage dashboards to visualize your data
New
Search for dashboards and folders
Filter by tag
Starred
Sort
Name
Tags
Monitoring
CPU & Memory
Node Exporter Full
linux
RPC

Скриншоты папок консоли:

```
red34@compute-vm-8-16-30-ssd-1772094666092: ~
red34@compute-vm-8-16-30-ssd-1772094666092:~$ ls /home/red34/red34/prometheus/
alerts.yml  prometheus.yml
red34@compute-vm-8-16-30-ssd-1772094666092:~$ cat /home/red34/red34/prometheus/prometheus.yml
global:
  scrape_interval: 15s

rule_files:
  - "alerts.yml"

scrape_configs:
  - job_name: 'prometheus'
    static_configs:
      - targets: ['localhost:9090']

  - job_name: 'node'
    static_configs:
      - targets: ['172.17.0.1:9100']

  - job_name: 'app'
    static_configs:
      - targets: ['192.168.240.23:8080']
    metrics_path: '/actuator/prometheus'
red34@compute-vm-8-16-30-ssd-1772094666092:~$ cat /home/red34/red34/prometheus/alerts.yml
groups:
  - name: cpu_alerts
    rules:
      - record: instance:cpu_usage:rate5m
        expr: rate(node_cpu_seconds_total{mode!="idle"}[5m])
      - alert: HighCPUUsage
        expr: instance:cpu_usage:rate5m > 0.8
        for: 1m
        labels:
          severity: warning
        annotations:
          summary: "Высокая загрузка CPU {{ $labels.instance }}"
red34@compute-vm-8-16-30-ssd-1772094666092:~$ ls /home/red34/red34/grafana/
docker-compose.yml  loki-config.yaml  promtail-config.yaml
red34@compute-vm-8-16-30-ssd-1772094666092:~$ cat /home/red34/red34/grafana/loki-config.yaml
auth_enabled: false

server:
  http_listen_port: 3100

common:
  path_prefix: /loki
  storage:
    filesystem:
      chunks_directory: /loki/chunks
      rules_directory: /loki/rules
  replication_factor: 1
  ring:
    kvstore:
      store: inmemory

schema_config:
  configs:
    - from: 2020-10-24
      store: tsdb
      object_store: filesystem
      schema: v13
      index:
        prefix: index_
        period: 24h
red34@compute-vm-8-16-30-ssd-1772094666092:~$ cat /home/red34/red34/grafana/promtail-config.yaml
server:
  http_listen_port: 9080
  grpc_listen_port: 0

positions:
  filename: /tmp/positions.yaml

clients:
```

red34@compute-vm-8-16-30-ssd-1772094666092: ~

```
  filename: /tmp/positions.yaml

clients:
  - url: http://loki:3100/loki/api/v1/push

scrape_configs:
  - job_name: containers
    docker_sd_configs:
      - host: unix:///var/run/docker.sock
        refresh_interval: 15s
    relabel_configs:
      - source_labels: ['__meta_docker_container_name']
        target_label: 'container'
      - source_labels: ['__meta_docker_container_log_stream']
        target_label: 'stream'
      - source_labels: ['__meta_docker_container_id']
        target_label: 'docker_id'
red34@compute-vm-8-16-30-ssd-1772094666092:~$ cat /home/red34/red34/grafana/docker-compose.yml
version: '3'

services:
  grafana:
    image: grafana/grafana:latest
    container_name: grafana34
    ports:
      - "3034:3000"
    environment:
      - GF_SECURITY_ADMIN_PASSWORD=admin
      - GF_FEATURE_TOGGLES_ENABLE=loki
    volumes:
      - grafana_data:/var/lib/grafana
    networks:
      - demo-network
    restart: unless-stopped

  loki:
    image: grafana/loki:latest
    container_name: loki34
    ports:
      - "3134:3100"
    command: -config.file=/etc/loki/local-config.yaml
    volumes:
      - ./loki-config.yaml:/etc/loki/local-config.yaml
      - loki_data:/loki
    networks:
      - demo-network
    restart: unless-stopped

  promtail:
    image: grafana/promtail:latest
    container_name: promtail34
    volumes:
      - ./promtail-config.yaml:/etc/promtail/config.yml
      - /var/run/docker.sock:/var/run/docker.sock
      - /var/lib/docker/containers:/var/lib/docker/containers:ro
      - /var/log:/var/log:ro
    command: -config.file=/etc/promtail/config.yml
    networks:
      - demo-network
    restart: unless-stopped
    depends_on:
      - loki

volumes:
  grafana_data:
  loki_data:

networks:
  demo-network:
    external: true
red34@compute-vm-8-16-30-ssd-1772094666092:~$
```

ENG 100% 17:29 27.03.2026